



Week 7 - Week 7 - Paleoclimate notes

Introduction to Climate Change (University of New South Wales)



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Week 7 – Lesson 1 – Paleoclimate – why do we care?

Lesson outcomes:

- Define paleoclimate
- Explain why it is important to understand how and why the climate as changed in the past
- Name two instances where climate played a role in changing human civilisation

What is paleoclimate?

Is the study of the Earth's past climates (climates that occurred before the instrumental record) is known as paleoclimate science.

Why do we study paleoclimate?

Attempting to make hypotheses about future climate is a serious challenge – we have one Earth, conducting experiments on it is not an option.

Components of the climate system operate and interact on a range of time scales; from fast (e.g., cloud formation) to the very slow (chemical weathering and tectonic plates).

- ➔ We need paleodata to understand how these slower components operate and interact.

Putting present day warming into context?

2021 was one of the seven warmest years on record.

2020 tied for the globally warmest year on record together with 2016.

The warmest years have all been since 2015.

RANDOM FACT: Number of stations in the International Surface Pressure Databank in 1859: 45 & 2010: 8869; we barely have 150 years of good quality data – reinforces the importance of studying paleoclimate.

But how can we be sure that the current trend is a symptom of anthropogenic forced climate change?

One way is to compare present day climate with natural climate variability over long periods of time.

- Temperatures the past 2000 years were relatively stable in comparison to rapid increase starting at the end of the 19th century.
- From CE900 to CE1200 – “Medieval Warm Period” – NH temp slightly warmer than millennial average.
- 1400s to the 1800s – “Little Ice Age” – NH temp slightly cooler.
- Warming observed since the late 1970s surpasses anything seen in proxy data since at least the beginning of the **Holocene (last 12,000 years)**.
- Preceding the Holocene – the Pleistocene or “Ice Age” – Earth's climate alternated between shorter warm (interglacial) periods and longer cold (glacial) periods.

- The Earth is currently in an interglacial – beginning around 12,000 years ago.

A changing landscape

As the Earth's climate changed, vegetation and animal life changed with it.

- E.g., 50m years ago – Svalbard (in the arctic circle next to Greenland) had a climate similar to Florida today – with crocodile like animals living in tropical rainforests.

The Last Ice Age reached its coldest point around 20,000 years ago (Last Glacial Maximum, or LGM):

Humans migrated northwards in Europe and many large land mammals disappeared all over the globe, driven to extinction by hunting and CC. Many human societies abandoned their hunter-gatherer lifestyles and elaborate forms of agriculture began to develop. The stable, warm and humid climate during the Holocene enabled the flourishing of human civilisation.

Has climate change affected humans in the past?

Even though Holocene climate was relatively stable, there have been civilisations which suffered, or even collapsed due to climate fluctuations.

E.g., the Aegean, Syro-Palestinian and Hittite civilisations all collapsed around 3200 years ago – Though to be in part due to a 300-year drought in the Eastern Mediterranean.

During the “Little Ice Age” (1400s – 1800s), a distinctly colder climate in Europe resulted in crop failures and famines, fuelling European migration to the Americas. Around this time, Viking settlements in Greenland declined and eventually disappeared. The cooling climate coincided with rising wheat prices, which peaked in 1816 – “year without summer” caused by Mt Tambora eruption.

Week 7 - Lesson 2 - Reconstructing past climates

Lesson outcomes:

- Define what a paleo proxy is
- Give examples of at least two proxies, how we use them and what they tell us
- Explain how isotopic analysis works

Introduction

We only have data for the last 150 years – the Earth has existed for 4.5 billion years.

To get an adequate perspective on climate change we must therefore reconstruct climate before the instrumental record – from indirect clues, known as proxy records or paleo proxies.

What is a paleoclimate proxy?

Paleoproxies are natural records that are influenced by climate.

They can be accessed by either looking at historical archives kept by people, or by examining and measuring physical or biological properties of natural archives (e.g. fossils, ice sheets, tree rings).

As proxies are often shaped by a combination of many factors, interpreting paleo proxies should be done very carefully.

Paleoproxies have two important characteristics: their span & resolution.

Span: how far back can a certain record go

Temporal Resolution: Using tree rings as an example – we might get information about whether a particular year was wet or dry – but no information about particular months, or days, hours or seconds in time.

So, the tree ring would have an annual resolution.

Modern meteorological instruments sometimes take measurements 50 times every second – much higher temporal resolution.

- Accurate dating can be a major challenge in paleoclimatic studies.
- Depending on the type of sample and its age, different dating methods are applied.

- These may be based on: decay rates of radioactive elements in the sample; time-dependent chemical changes in the sample; correlation with past reversals of the Earth's magnetic field; growth rates of biological organisms.
- The ages of samples may also be calibrated with changes in the Earth's orbit or correlated with each other using Stratigraphic methods.

Historical records

Some examples of human formulated records kept over several hundred years are: dates for grape harvest in France; frequency of major dust storms in China; height of the Nile River in Cairo; number of weeks during which sea ice reached the coast of Iceland; dates for the blooming of cherry flowers in Kyoto; dates for freezing of Lake Suwa; wheat prices in Europe.

In addition, anecdotal information may be used, e.g., from diaries, old newspapers.

Historical records are useful for reconstructing climate for the last 1000 years – but to reconstruct climate before that time, it is necessary to access information stored in natural archives.

Tree rings

- Tree growth: highly influenced by climate
- Cross section from intermediate and high latitude regions reveals lighter and darker bands.
- Differences in thickness and density of these annual layers reflects temperature, rain, sunshine, wind speed.

An example

- During a warm (not too hot) growing season with ample rainfall – growth rings will be wide
- Water scarce growing season or extreme temperatures – growth is diminished.

Advantages of tree ring records

- Trees grow a new layer each year – so, if you know when a tree existed – the tree rings give you annual records of the climate for that region.

Downsides of tree ring records

- They only extend back a few thousand years at most.
- However, fossilised trees that lived tens of thousands of years ago can be dug up in certain regions & can be carbon dated.

Corals

- Corals extract carbonate and calcium ions from surrounding seawater – to build their aragonite skeletons.
- These skeletons form annual bands which contain isotopes and trace metals that provide information about temperature and composition of the water in which they grow.

Uses

- **Warm water** corals only grow in shallow water – thus can be used as proxies for surface water temperatures, rainfall, and water clarity, as well as changes in sea level.
- Benefit of annual layers
- Corals can live for several hundred years
- Dead coral deposits – can provide record of several thousand years.
- **Cold water corals**, some of which grow uninterrupted for thousands of years – give information on the conditions of the deep ocean over time.

Ice cores

- Provide data up to hundreds of thousands of years old.
- Longest continuous record recovered – spans 800,000 years
- Scientists are trying to find chunks of ice that might be much older (non-continuous)
- 2017 example of 2.7-million-year-old ice sample.

What scientists measure

- Oxygen isotope composition – used as a proxy for past temperatures
- Air bubbles – samples of atmospheric gases
- Thickness of ice layers – info about local precipitation.
- Airborne particles – e.g., dust or volcanic ash.

Sediments

- Composed of biogenic and land-derived materials and accumulate at the bottom of oceans and lakes.
- Layers preserve some of the best and longest records of past climate variations – up to hundreds of millions of years – although their resolution is typically not higher than a decade
- Foraminifera fossils – microorganisms – surface dwelling (planktic) – sea floor dwelling (benthic)
- Pollen grains
- <https://www.youtube.com/watch?v=oaOfesJZ3lY&t=2s>

Isotopic analysis

Isotopes are variants of atoms that have the same number of protons but different number of neutrons, hence a different mass.

Isotopes can be stable or unstable (radioactive)

Variations in the relative abundance of a stable isotope often reflects changing environmental conditions, hence some isotopes (e.g., oxygen, carbon, hydrogen, nitrogen) are used as climate proxies.

The processes that affect stable isotope ratios are called isotope fractionation.

Isotope ratios preserved in ancient rocks, fossils, or ice contains information about the climate in which these structures formed. Three examples of stable isotopes are ^{18}O , ^2H and ^{13}C .

^{18}O Isotope:

Most oxygen atoms occur as the ^{16}O isotope, while about 1 in 500 occurs as the heavier ^{18}O . The lighter water molecule H_2^{16}O is more likely to evaporate and less likely to condensate than the heavier H_2^{18}O . As a result, it is more likely to be transported by the atmosphere to polar regions where some of it snows out over accumulating ice sheets. During cold periods, more of the ^{16}O is therefore stored in ice sheets, and hence the ocean becomes enriched in ^{18}O . As ice sheets melt during deglaciations, they release ^{16}O back into the ocean. As shown in the video above (sediments), this process allows us to reconstruct the total ice volume on land based on marine sediment records.

Week 7 - Lesson 3 - A variable climate: lessons from the past

Climate change: speed matters!

The carbon cycle operates on timescales of years to many millions of years.

Fast end of time scale: carbon can be taken up and released by vegetation and soil.

Slightly longer end of time scale: absorption of carbon by the oceans via the solubility and biological pumps.

Long end of time scale: complex processes of chemical weathering which, in combination with changes in ocean chemistry, can alter atmospheric CO_2 levels over millions of years.

Glacial cycles

Let's define glacial and ice-age.

We are technically in an ice age right now – that started 2.6 million years ago.

Ice-age: period of time where the Earth has land-based ice sheets (e.g. Greenland and Antarctica).

Glacial periods: are times during our present ice-age when large ice sheets were present in the Northern Hemisphere, over North America and Eurasia.

The current ice age has seen many glacial episodes.

The time between two glacials: inter-glacials

During interglacials, North America and Eurasia do not support any major ice sheets (today 4 example)

During the past 1m years, glacial-interglacial cycles occurred at a frequency of 100,000 years.

Need to come back and finish this lesson.